1. Initial Setup & Configuration

Before tracking files, RStudio must know where Git is located on your computer and who you are. [1, 2]

1. **Link Git to RStudio:**
    - Go to **Tools** > **Global Options...** > **Git/SVN**.
    - Ensure **Enable version control interface for RStudio projects** is checked.
    - If the **Git executable** path is blank or shows `(none)`, click **Browse** and point it to your Git installation (e.g., `/usr/bin/git` on Mac or `C:/Program Files/Git/bin/git.exe` on Windows). Restart RStudio if prompted. [1, 2, 3]
2. **Introduce Yourself to Git:**

You can configure your Git identity effortlessly inside R without using the command line by installing the `usethis` library. Run the following code in your R console:

R

```
install.packages("usethis")
usethis::use_git_config(user.name = "Your Name", user.email =
"your_email@example.com")
```

    - Use code with caution.

    (Note: Use the exact email address tied to your GitHub account). [1, 2]

2. Connect a Project to Git

Git tracks files at the **RStudio Project** level. You have two main routes depending on whether your project already exists: [1]

Route A: Start Fresh by Cloning from GitHub (Recommended)

1. Go to GitHub, create a new repository, and check **Add a README file**. [1]
2. Click the green **Code** button and copy the repository URL. [1]
3. Inside RStudio, go to **File** > **New Project...** > **Version Control** > **Git**. [1, 2]
4. Paste the URL into **Repository URL**, choose a local directory to save it, and click **Create Project**. [1, 2]

Route B: Turn an Existing Local Project into a Git Repo

1. Open your existing project in RStudio.
2. Go to **Tools** > **Project Options...** > **Git/SVN**.
3. Change the Version Control System from *None* to **Git**, and click OK.
4. Click **Yes** to restart RStudio when prompted. [1, 2]

3. The Daily Git Workflow

Once active, a dedicated **Git** tab will appear in your environment/history pane (usually top-right). [1, 2]

```
[ Modify Files ] ──> [ Stage ] ──> [ Commit (Save Locally) ] ──> [ Push (Upload to GitHub) ]
```

1. **Make Changes:** Create or edit your R scripts or R Markdown documents and save them as usual. [1]
2. **Stage:** Open the **Git** tab. You will see a list of modified or untracked files. Check the **Staged** box next to the files you want to save in this version snapshot. [1, 2]
3. **Commit:** Click the **Commit** button on the Git toolbar. This opens a staging window. Type a concise, descriptive **Commit Message** (e.g., *"Fix summary statistics function"*) and click **Commit** to save these changes locally. [1, 2]
4. **Push / Pull:**
    - Click the green **Push** (Up Arrow) button to upload your local snapshots online to GitHub.
    - Click the blue **Pull** (Down Arrow) button before you start working to download any changes made by collaborators or from another device. [1, 2]